

Topic: The Impact of Remote-Working on Cybersecurity Risks and Defence Strategies in Fintech Organisations.

Module: Contemporary Issues in Computer Science.

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Assessment 1: Literature Review

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An AI Use Declaration is included in Appendix A

1. Introduction

Financial technology has become central to modern finance, with fintech organisations delivering services through digital banking, mobile payments, peer-to-peer lending, robo-advisory platforms, blockchain and online payment infrastructures (Ali et al., 2024). Many fintech firms are digital-first, relying on cloud computing, mobile applications, APIs and third-party integrations, which supports innovation but widens the cybersecurity attack surface because sensitive customer data and high-value transactions are processed online (Pittala and Kacheru, 2025; Onwuegbuchi et al., 2023). Remote and hybrid working have further changed this risk environment, as decentralised work models and flexible digital collaboration are now more embedded across financial and cybersecurity sectors (Azeem Mohammed, Rakibuzzaman and Ashraful Alam, 2025). Employees may access systems through home networks, personal devices and cloud services, weakening traditional security boundaries (Odeh and Odeh, 2025). Therefore, this review evaluates cybersecurity risks and mitigation strategies in fintech, including resilience, compliance and operational security (Basak and Tiwari, 2025; Khaliq, Sultan and Zafar, 2025; Karakasilioti, 2024).

2. Research Question and Objectives

Research Question: To what extent has remote working increased cybersecurity risks in fintech organisations, and how effective are current mitigation strategies?

Objectives:

1. To analyse the cybersecurity risks introduced by remote working in fintech organisations.
2. To evaluate the effectiveness of mitigation strategies such as multi-factor authentication, VPNs, and continuous monitoring systems within the fintech context.
3. To identify gaps in current research and propose improvements for cybersecurity practices in fintech organisations.

3. Literature Review

Theme 1: Cybersecurity Risks Introduced by Remote Working in Fintech

Remote working has increased cybersecurity risks in fintech organisations because it extends access to sensitive financial systems beyond controlled office environments. Fintech operates in a high-risk ecosystem built around online platforms, mobile banking, digital payments and technology-driven services. Kamuangu (2024) identifies data breaches, phishing, malware, compliance and authentication as major concerns, showing cybersecurity is central to trust in digital services. However, this source mainly explains the fintech threat landscape, so it must be combined with remote-work research to show how working from home and hybrid work intensifies risks.

A major risk is personal devices and unmanaged endpoints. Chigada and Daniels (2021) argue that BYOD practices expose financial services firms to data breaches, unauthorised access and security weaknesses when personal devices connect to corporate infrastructure without effective monitoring. This is relevant because fintech organisations process customer financial data. However, the study involved eight participants, limiting its generalisability across fintech organisations with different security maturity levels.

Rah (2023) supports this concern by showing that BYOD security in remote workplaces depends on controls such as device management, encryption, authentication, VPNs, antivirus, updates and patches. The study suggests remote-working risk is produced by the interaction between personal devices, user behaviour and inconsistent controls. However, because the study is not fintech-specific, its findings must be applied carefully to organisations with stricter data-protection requirements.

Home-office environments create risks that differ from office settings. Klint (2023) identifies phishing exposure, VPN weaknesses, outdated devices, insecure routers and physical security concerns. Scott (2022) adds that working from home can increase phishing susceptibility through isolation, workload, time pressure and emotional states, showing that phishing is technical and human. Angafor, Yevseyeva and Maglaras (2024) similarly argue that technical controls provide partial protection if users lack security awareness.

Remote access and insider threats create further risk. Vitla (2024) identifies unsecured RDP access as a vulnerability enabling unauthorised access, ransomware and data exfiltration. Ogundipe (2024) argues that insider threats are difficult in financial institutions because employees and contractors already have legitimate access to sensitive systems. In remote fintech environments, reduced supervision may make misuse harder.

Overall, remote working widens fintech's attack surface across devices, home networks, remote access tools, phishing and insider access. However, the limited research combining remote working, cybersecurity and fintech should not be interpreted as evidence that the issue is unimportant. Existing literature often examines fintech cybersecurity and remote-working risks separately, leaving a clear gap around how remote work reshapes fintech-specific risk.

Theme 2: Effectiveness of Mitigation Strategies in the Fintech Context

Fintech organisations use layered controls to reduce remote working risks, but effectiveness depends on implementation rather than the control itself. Multi-factor authentication is promoted because remote employees and customers access sensitive systems outside the perimeter. Alsumayt et al. (2025) argue that MFA strengthens access security through identity checks and supports confidentiality and integrity. Although their focus is operational technology, the principle is relevant to fintech because weak authentication could expose payment platforms and customer accounts. However, MFA can be difficult to integrate where legacy systems remain priorities.

Recent MFA research complicates the assumption that MFA solves credential risk. Carjuman, Fook and Hlaing (2025) argue that static MFA based on passwords and one-time codes is vulnerable to phishing, SIM swapping, session hijacking and man-in-the-middle attacks. Their framework combines biometric verification, device fingerprinting and contextual risk scoring to increase security while reducing friction. Authentication must protect accounts without damaging customer experience. However, the framework is proposed rather than tested in live fintech environments, so its effectiveness remains theoretical.

VPNs are central to remote access security but should be treated as necessary rather than sufficient. Tekniska et al. (2023) found that remote work increased VPN use, network complexity and attack surface. Their review shows that VPN vulnerabilities were exploited, including by advanced persistent threat groups, and recommends MFA, patching and layered defences. This suggests that VPNs protect data in transit but do not address endpoint compromise, poor authentication or insecure configuration.

Zohaib et al. (2024) extend this point through Zero Trust VPN, arguing that traditional VPNs raise security and scalability concerns in hybrid work environments. Their framework combines VPN principles with Zero Trust Network Access, SASE and related technologies. This is relevant to fintech because cloud-native and API-driven services require access decisions based on identity, device posture and risk rather than network connection. However, fintech-specific evidence remains limited.

Continuous monitoring and SOC capability form another mitigation layer. Khan et al. (2024) explain that SIEM systems collect and analyse security events, but prioritisation becomes difficult as alert volumes increase. Their machine-learning approach uses prediction probability, impact score, mean time to detect and mean time to resolve to support prioritisation. This is useful for fintech SOC teams, where delayed detection could affect payments, fraud prevention and trust. However, the study does not directly evaluate remote fintech SOC environments.

Madat, Poriya and Srivaramangai (2025) argue that traditional office-based security tools are inadequate for remote work and recommend stronger access control, real-time threat detection, cryptography, incident response and endpoint protection. This supports defence-in-depth. Nevertheless, many studies examine controls separately, while fintech organisations need evidence on how MFA, VPNs, Zero Trust, SIEM, endpoint security and training operate together.

Theme 3: Gaps in the Literature

Several gaps limit the evidence available to fintech organisations. First, fintech is absorbed into broader financial services research, even though it differs from traditional banks through cloud-native systems, APIs and leaner security governance.

Umoga et al. (2024) identify phishing, malware, ransomware, supply chain attacks and AI-driven threats as major fintech risks. However, this literature reviews fintech threats generally rather than analysing how remote working reshapes risk for employees and SOC teams. This shows a gap, as studies often discuss fintech cybersecurity and remote-working cybersecurity separately.

Second, cybersecurity is not always treated as socio-technical. Mahmood, Chadhar and Firmin (2024) argue that crisis-related challenges should be understood through social, technical, political, economic and environmental subsystems. This is valuable because remote fintech security depends on technical tools, employee behaviour, regulation and culture. However, their research focuses on higher education, so the framework requires fintech testing.

Third, human factors remain underdeveloped in fintech-focused research. Bishop (2025) emphasises that cybersecurity culture includes leadership, policies, awareness, communication, compliance and improvement. This matters because remote workers make security decisions without direct supervision. Yet cybersecurity culture is usually discussed generally, not in relation to fintech pressures such as speed and accountability.

Finally, the SOC perspective remains weak. Sohal (2025) links SOC maturity to improved incident response through threat intelligence, detection engineering, adversary emulation and automation. However, little research examines how remote work affects SOC collaboration, alert fatigue and response times in fintech. Future research should examine remote fintech cybersecurity as an integrated system where authentication, endpoint security, monitoring, culture and SOC effectiveness intersect.

4. Implications for Research and Practice

This review suggests that fintech organisations should not treat remote-work security as a checklist of separate tools. MFA, VPNs, SIEM, endpoint protection and awareness training are most useful when combined as a defence-in-depth approach. This matters because remote fintech work involves sensitive customer data, cloud platforms, APIs and transactions being accessed from less controlled environments. Pomerleau and Maimon (2022) argue that cybersecurity practice should be evidence-

based, as organisations often invest in tools without clear proof that they reduce risk in real settings. For fintech practice, this means security teams should measure whether controls reduce phishing success, unauthorised access, mean time to detect and mean time to respond, rather than assuming that deployment equals protection. For research, the main priority is to study remote fintech cybersecurity as a socio-technical system, where technology, employee behaviour, culture and SOC effectiveness interact.

5. Reflection

This review developed my understanding of remote-work cybersecurity in fintech. At first, I assumed that common controls such as MFA, VPNs and staff training were sufficient to manage most risks. However, the literature showed that effectiveness depends on implementation quality, organisational culture, user behaviour and monitoring capability. I also learned that security weaknesses can result from small design or testing failures, not only obvious attacks. Aljaidi et al. (2022) demonstrate this through the iTunes updater case, where an unquoted file path and trusted software behaviour could allow ransomware execution. Although this example is not fintech-specific, it reinforces the importance of secure endpoint configuration, patching and quality assurance in remote-working environments. The most important lesson from this review is that cybersecurity should be evaluated critically: not only by asking whether a control exists, but whether it works in context and is supported by evidence. This will shape my Assessment 2 proposal by focusing on SOC effectiveness in hybrid fintech environments.

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Appendices

Appendix A

ChatGPT was used as an academic support tool during the preparation of this assessment. It was used to support brainstorming, planning, explanation of concepts, report structure, search phrase development, proofreading and presentation preparation.

All AI-assisted content was reviewed, edited and adapted by me. The final topic, research question, objectives, argument, source selection, critical evaluation and submitted work reflect my own understanding and academic judgement. AI was not used to fabricate sources, create false references, collect primary data or replace my own analysis.